Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Sheikh.
21/02/26

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Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT**
Semester	**6**	**Course Code**	**23IOT1623**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

TA Activity	**Demonstration Based Learning**
Course Outcome	**CO1** To understand the different flavors of mobile operating system and their specific features. **CO2** Apply Android applications using various UI controls and layouts. **CO3** Analyze applications capable of data handling and communication with native components. **CO4** Create and develop applications that support multiple devices and configurations.
Problem Definition	1. Develop a Student Registration Form using Kotlin in Android Studio. 2. Implement appropriate layout types (Linear Layout / Relative Layout / Constraint Layout). 3. Apply event handling and basic validation. 4. Demonstrate the working application during evaluation. 5. Generate a detailed report based on your implementation and submit it for evaluation (hard copies & soft copies both)
Theory (100 words)	The **Student Registration App** on GitHub is an Android project developed in Kotlin using Android Studio. It implements a three-activity flow: the main screen with a register button, a registration form with fields for name, email, password, phone, roll number, city, and gender, and a result screen that displays submitted details. Layouts use Material components, CardViews, ConstraintLayout, and input validation provides user feedback. Data is passed between activities using Intents. The UI is enhanced with modern design elements like gradient backgrounds and floating TextInputLayouts. This project demonstrates event handling, form validation, activity navigation, and enhanced UI design.
Procedure and Execution (100 Words)	Step for Implementation: ➢ Open Android Studio and create a new project using **Empty Views Activity** with Kotlin as the programming language. ➢ Design the MainActivity layout with a button to navigate to the registration form. ➢ Create RegisterActivity and design the registration form using ConstraintLayout, CardView, ScrollView, and Material TextInputLayout components.

- Add input fields for name, email, password, phone number, roll number, city, and gender selection using RadioGroup.
- Implement basic validation in Kotlin to check empty fields, valid email format, and correct phone number length.
- Create ResultActivity to display the entered details.
- Pass data between activities using Intent.

- Run the application on an emulator or Android device.
- The main screen appears with a Register button.
- Clicking the button opens the registration form.
- The user enters details and clicks Register.
- If validation is successful, the data is displayed on the result screen.

Code:

1. MainActivity.kt

```
package com.example.studentregistration

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val btnRegister = findViewById<Button>(R.id.btnRegister)

        btnRegister.setOnClickListener {
            val intent = Intent(this, RegisterActivity::class.java)
            startActivity(intent)
            overridePendingTransition(android.R.anim.slide_in_left,
android.R.anim.slide_out_right)
        }
```

```
    }
}
```

2. RegisterActivity.kt

```
package com.example.studentregistration

import android.content.Intent
import android.os.Bundle
import android.util.Patterns
import android.widget.*
import androidx.appcompat.app.AppCompatActivity

class RegisterActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_register)

        val etName = findViewById<EditText>(R.id.etName)
        val etEmail = findViewById<EditText>(R.id.etEmail)
        val etPassword = findViewById<EditText>(R.id.etPassword)
        val etPhone = findViewById<EditText>(R.id.etPhone)
        val etRoll = findViewById<EditText>(R.id.etRoll)
        val etCity = findViewById<EditText>(R.id.etCity)
        val radioGroup = findViewById<RadioGroup>(R.id.radioGroup)
        val btnSubmit = findViewById<Button>(R.id.btnSubmit)

        btnSubmit.setOnClickListener {

            val name = etName.text.toString().trim()
            val email = etEmail.text.toString().trim()
            val password = etPassword.text.toString().trim()
            val phone = etPhone.text.toString().trim()
            val roll = etRoll.text.toString().trim()
            val city = etCity.text.toString().trim()

            val selectedGenderId = radioGroup.checkedRadioButtonId

            if (name.isEmpty()) {
                etName.error = "Name required"
                return@setOnClickListener
            }
```

```
        if (!Patterns.EMAIL_ADDRESS.matcher(email).matches()) {
            etEmail.error = "Valid email required"
            return@setOnClickListener
        }

        if (password.length < 6) {
            etPassword.error = "Password must be at least 6 characters"
            return@setOnClickListener
        }

        if (phone.length != 10) {
            etPhone.error = "Enter 10 digit number"
            return@setOnClickListener
        }

        if (roll.isEmpty()) {
            etRoll.error = "Roll number required"
            return@setOnClickListener
        }

        if (city.isEmpty()) {
            etCity.error = "City required"
            return@setOnClickListener
        }

        if (selectedGenderId == -1) {
            Toast.makeText(this, "Select Gender",
Toast.LENGTH_SHORT).show()
            return@setOnClickListener
        }

        val gender =
findViewById<RadioButton>(selectedGenderId).text.toString()

        val intent = Intent(this, ResultActivity::class.java)
        intent.putExtra("name", name)
        intent.putExtra("email", email)
        intent.putExtra("phone", phone)
        intent.putExtra("roll", roll)
        intent.putExtra("city", city)
        intent.putExtra("gender", gender)

        startActivity(intent)
```

```
            overridePendingTransition(android.R.anim.slide_in_left,
android.R.anim.slide_out_right)
        }
    }
}
```

3. ResultActivity.kt

```
package com.example.studentregistration

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class ResultActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_result)

        val tvDetails = findViewById<TextView>(R.id.tvDetails)

        val name = intent.getStringExtra("name")
        val email = intent.getStringExtra("email")
        val phone = intent.getStringExtra("phone")
        val roll = intent.getStringExtra("roll")
        val city = intent.getStringExtra("city")
        val gender = intent.getStringExtra("gender")

        tvDetails.text = """
            Name: $name
            Email: $email
            Roll No: $roll
            Phone: $phone
            City: $city
            Gender: $gender
        """.trimIndent()
    }
}
```

4. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F7F7F9">

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <TextView
            android:id="@+id/tvName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Sheikh Wasimuddin"
            android:textSize="28sp"
            android:textStyle="bold"
            android:textColor="#2C2C2C"
            android:letterSpacing="0.03"
            android:layout_marginBottom="30dp"/>

        <Button
            android:id="@+id/btnRegister"
            android:layout_width="220dp"
            android:layout_height="55dp"
            android:text="Register"
            android:textAllCaps="false"
            android:textSize="16sp"
            android:textColor="@android:color/white"
            android:backgroundTint="#3A4F7A"
```

```
        android:elevation="6dp"/>

    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

5. activity_register.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F7F7F9">

    <androidx.cardview.widget.CardView
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_margin="24dp"
        app:cardCornerRadius="24dp"
        app:cardElevation="8dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <ScrollView
            android:layout_width="match_parent"
            android:layout_height="wrap_content">

            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="vertical"
                android:padding="24dp">

                <TextView
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Student Registration"
                    android:textSize="20sp"
                    android:textStyle="bold"
                    android:textColor="#3A4F7A"
```

```
        android:layout_marginBottom="20dp"/>

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"
        android:layout_marginTop="15dp"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="15dp"/>

    <EditText
        android:id="@+id/etPhone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone Number"
        android:inputType="phone"
        android:layout_marginTop="15dp"/>

    <EditText
        android:id="@+id/etRoll"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Roll Number"
        android:layout_marginTop="15dp"/>

    <EditText
        android:id="@+id/etCity"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter City"
```

```
        android:layout_marginTop="15dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Gender"
        android:layout_marginTop="20dp"/>

    <RadioGroup
        android:id="@+id/radioGroup"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">

        <RadioButton
            android:id="@+id/rbMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male"/>

        <RadioButton
            android:id="@+id/rbFemale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Female"/>
    </RadioGroup>

    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:layout_marginTop="25dp"/>

        </LinearLayout>

    </ScrollView>

  </androidx.cardview.widget.CardView>

</androidx.constraintlayout.widget.ConstraintLayout>
```

6. activity_result.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#F7F7F9">

    <androidx.cardview.widget.CardView
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_margin="24dp"
        app:cardCornerRadius="24dp"
        app:cardElevation="10dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent">

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:padding="28dp"
            android:gravity="center">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Registration Summary"
                android:textSize="22sp"
                android:textStyle="bold"
                android:textColor="#3A4F7A"
                android:layout_marginBottom="18dp"/>

            <TextView
                android:id="@+id/tvDetails"
                android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
            android:textSize="16sp"
            android:textColor="#2C2C2C"
            android:lineSpacingExtra="6dp"
            android:textAlignment="center"/>

        </LinearLayout>

    </androidx.cardview.widget.CardView>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:

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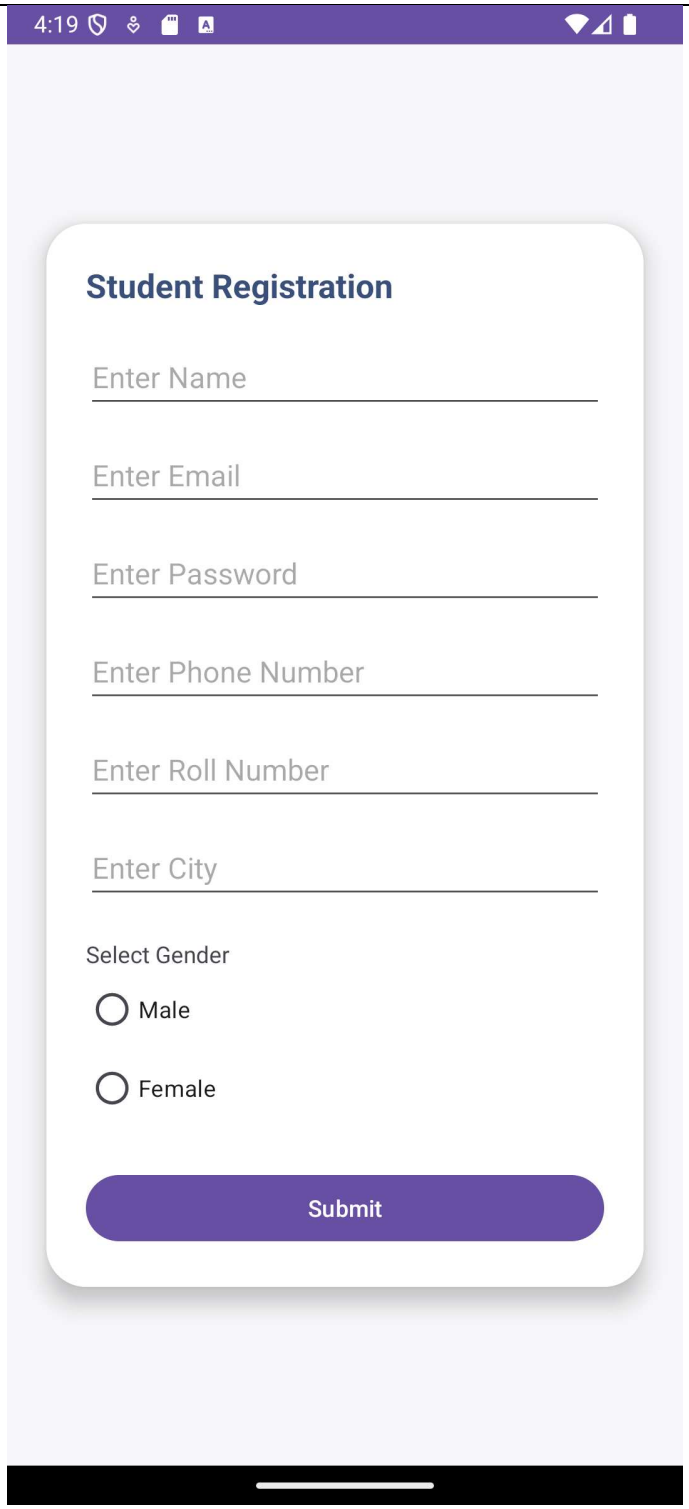

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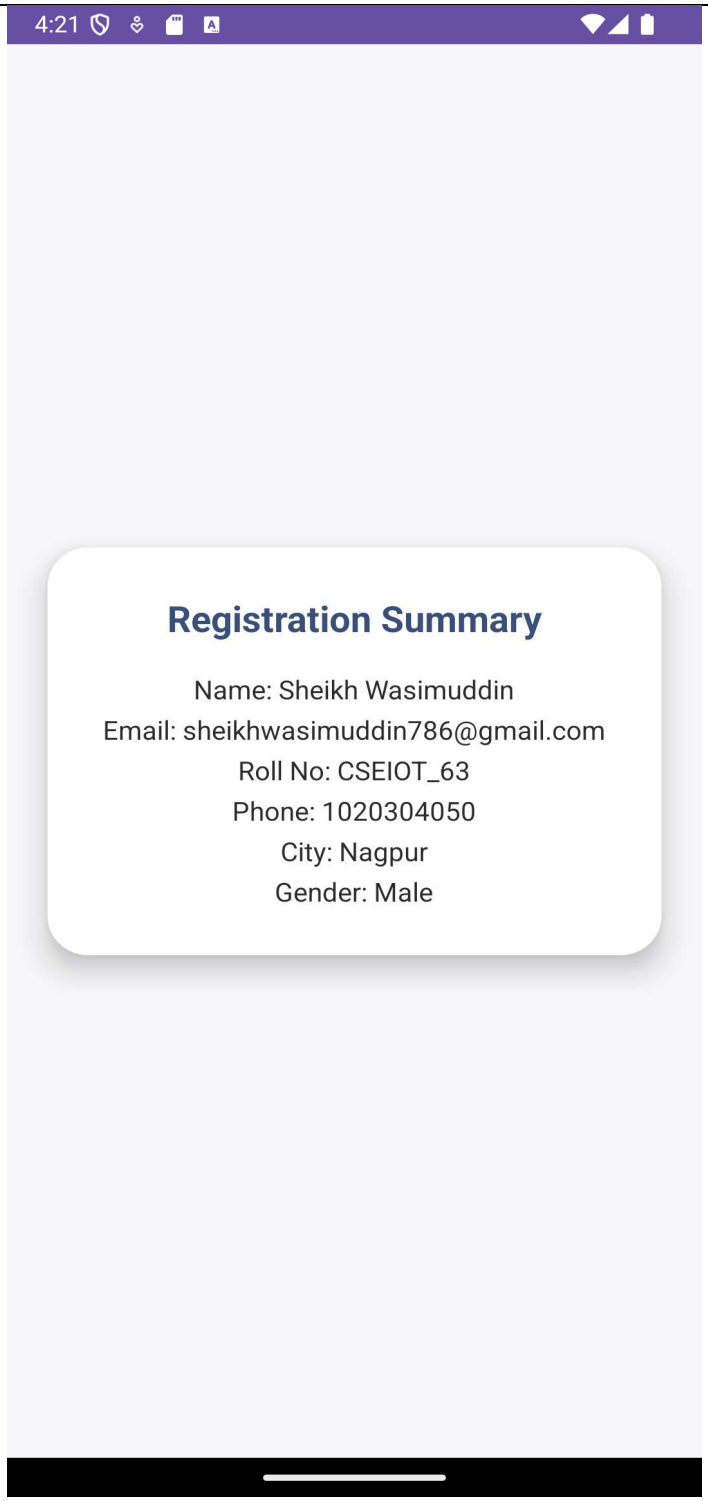

Output Analysis	The application executes successfully and displays three functional screens demonstrating proper activity navigation and UI implementation. **1. Main Screen (Home Activity)** The first screen displays the developer's name, "Sheikh Wasimuddin," at the center along with a styled Register button. The layout is clean and centered using ConstraintLayout. When the user clicks the Register button, the application navigates to the registration form screen using Intent. This confirms proper event handling and activity transition. **2. Registration Form Screen** The second screen displays a well-designed Student Registration form inside a CardView with elevation and rounded corners. The form contains input fields for Name, Email, Password, Phone Number, Roll Number, City, and Gender selection using RadioButtons. Material TextInputLayouts provide floating labels and structured input fields. The application performs validation: • Empty fields show error messages. • Email format is validated. • Phone number length is checked. • Gender must be selected. Only after successful validation does the form proceed to the next screen. This confirms correct implementation of input validation and user interaction handling. **3. Registration Summary Screen (Result Activity)** The third screen displays a summary of the entered details inside a styled CardView. The data is passed from RegisterActivity to ResultActivity using Intent. All user-entered information is displayed correctly, confirming proper data transfer between activities.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin/Student-Registration-App
Conclusion	The Student Registration App was successfully developed using Kotlin in Android Studio. The application demonstrates the implementation of

	multiple activities, user interface design, event handling, and form validation. The use of ConstraintLayout, CardView, and Material components enhances the visual appearance and usability of the application. Input validation ensures accurate data entry, improving reliability and user experience. Data is effectively transferred between activities using Intent, confirming proper activity navigation. The project fulfills all the required objectives and showcases practical knowledge of Android application development concepts, including UI design, validation logic, and activity lifecycle management
Plag Report (Similarity index < 12%)	
Date	**21/02/2026**